Deep Learning: Mock Exam (90 min)

[short] Learning XOR

Frage 1 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

You are given two Neural Network architectures:

```
class Net1(nn.Module):

    def __init__(self):
        super().__init__()
        self.layers = nn.Sequential(nn.Linear(2, 1))

    def forward(self, x):
        return self.layers(x)
```

```
class Net2(nn.Module):

    def __init__(self):
        super().__init__()
        self.layers = nn.Sequential(nn.Linear(2, 10), nn.Linear(10, 1))

    def forward(self, x):
        return self.layers(x)
```

Which of these, if any, can theoretically learn the XOR function?

- ○ Net2
- ○ None
- ○ Net1
- ○ Both

[short] Depthwise Separable Convolutions

Frage 2 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Construct a true statement:

Compared to regular convolutions, *depthwise separable convolutions* have [--- bitte auswählen ----] parameters, [--- bitte auswählen ----] computational costs, and [--- bitte auswählen ----] representational capacity.

(grading note: 1/3pt per correctly selected option)

1) fewer / more 2) lower / higher / similar 3) lower / higher

[short] Cross-Entropy and Accuracy

Frage 3 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Which of the following statements about the relationship that exists between the cross-entropy loss (CE) and accuracy (AC) is/are correct?

- [ ] CE = 0 implies AC = 100%
- [ ] lower CE implies higher AC
- [ ] higher CE implies higher AC
- [ ] AC = 100% implies CE = 0

Numpy instead of for-loops

Frage 4 von 58 (2 Punkte)
Nicht beantwortet

Aktionen

Complete the python script below

```
import numpy as np

N = 6
D = 4
M = 8

X = np.random.rand(N, D)
W = np.random.rand(D, M)
b = np.full((M,), 0.1)

def linear_python(X, W, b):
    y = []
    for i in range(N):
        y_i = []
        for j in range(M):
            y_ij = b[j]
            for k in range(D):
                y_ij += X[i, k] * W[k, j]
            y_i.append(y_ij)
        y.append(y_i)
    return np.array(y)


def linear_numpy(X, W, b):
    # START TODO #################
    # Re-implement linear_python without for loops using Numpy
    raise NotImplementedError
    # END TODO #################
    return y
```

```
assert np.allclose(linear_python(X, W, b), linear_numpy(X, W, b))
```

In the textbox below, write the python code that implements the linear_numpy function such that it

- **is functionally equivalent to linear_python**
- **uses no for loops**
- **uses proper syntax! (after adding your code to the todo block, the script above should execute without exceptions)**

P

[short] Padding in Convolutions

Frage 5 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Answer the two questions below (0.5p per correct answer)

a) When using "valid convolution", what padding is used in a 5x5 convolution with stride 1?

Answer:

b) When using "same convolution", what padding is used in a 5x5 convolution with stride 1?

Answer:

[short] Combining Data Augmentations

Frage 6 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Select all that apply: Which of the following statements is/are generally true?

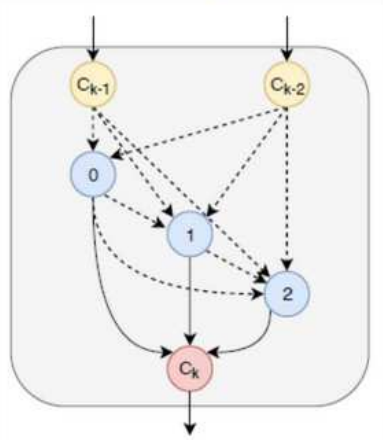
- [ ] AutoAugment samples a random augmentation and its strength for each batch
- [ ] RandAugment samples a random augmentation and its strength for each batch
- [ ] TrivialAugment and RandAugment sample random augmentations
- [ ] AutoAugment optimizes the combination of augmentations for a model
- [ ] TrivialAugment always randomly chooses between applying Rotation or ColorJitter to each batch
- [ ] TrivialAugment samples a single augmentation for each image in a batch

One-shot Model

Frage 7 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Consider the following one-shot model:



Each of the dashed edges can take one of the following operations:

- **1x1 conv**
- **3x3 conv**
- **3x3 avgpool**
- **skip connection**
- **identity**

How many different architectures does the one-shot model search space span?

Simply count the number of possible assignments of operations to edges here and ignore the fact that e.g. symmetry makes some of these architectures equivalent.

[short] Statements about Batch Size

Frage 8 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Which of the following statement(s) is(are) correct?

- [ ] Mini-batches can reduce the memory cost, but cannot impact the final performance
- [ ] Small batch sizes can have a regularizing effect
- [ ] For smaller batch sizes, the optimal learning rate is typically smaller
- [ ] We should always use an as large a batch size as possible

[short] Main Challenges in Deep Learning

Frage 9 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Which are the main challenges in optimizing deep neural networks?

- [ ] Ill-conditioning
- [ ] Existence of many poor local optima
- [ ] Large datasets
- [ ] The lack of saddle points
- [ ] High number of trainable parameters of neural networks

[short] Speeding up NAS

Frage 10 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Which of the following techniques can be used to speed up NAS?

- [ ] Using RL to design the cells
- [ ] Training the models on smaller datasets
- [ ] Using a blackbox optimizer
- [ ] Inheriting weights using morphisms

[short] LSTM Cells

Frage 11 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Construct a correct statement:

In an LSTM cell, if the output value of the ---- bitte auswählen ---- gate is close to ---- bitte auswählen ----, information from the previous cell state will be retained.

1) forget / input / output 4) -1 / 0 / 1

[short] BBO HPO Methods

Frage 12 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Which of the following statements is/are true?

- [ ] Random Search never outperforms Hyperband
- [ ] Successive Halving risks dropping good configurations early
- [ ] BOHB uses Gaussian Processes to select configurations
- [ ] Hyperband uses Random Search to select configurations

[short] RNN: LSTM

Frage 13 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

LSTMs are better than vanilla RNNs in what sense?

- [ ] They are easier to train
- [ ] They can more easily forget information that is not useful
- [ ] They do not have recurrence
- [ ] They can more easily model longer-term dependencies in the sequence

[short] MC Drop-out

Frage 14 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Which of the following statements is/are true about MC drop-out:

- [ ] Inference in MC drop-out is more efficient than in regular drop-out
- [ ] Using MC drop-out, units are dropped at test time
- [ ] Using MC drop-out, units are dropped at training time
- [ ] Training in MC drop-out is more efficient than in regular drop out

Input and Output Dimensions

Frage 15 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Given the following network class:

```
import torch.nn as nn
import torch.nn.functional as F


class Net1(nn.Module):

    def __init__(self):
        super().__init__()
        self.conv1 = nn.Conv2d(3, 6, 5)
        self.pool = nn.MaxPool2d(2, 2)
        self.conv2 = nn.Conv2d(6, 16, 5)
        self.fc1 = nn.Linear(16 * 5 * 5, 20)
        self.fc2 = nn.Linear(20, 10)

    def forward(self, x):
        x = self.pool(F.relu(self.conv1(x)))
        x = self.pool(F.relu(self.conv2(x)))
        x = x.view(-1, 16 * 5 * 5)
        x = F.relu(self.fc1(x))
        x = self.fc2(x)
        return x
```

Which of the following statements is/are true?

- [ ] Using a batch size of 8, Net1 produces outputs of size [8, 20]
- [ ] Using a batch size of 4, Net1 produces outputs of size [4, 10]
- [ ] Net1 can process RGB images of size 32 x 32
- [ ] Net1 can process RGB images of size 28 x 28

BLR in 1D

Frage 16 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Consider the function $f(x) = w\,x$ of a single variable x, which represents a line with slope w and no intercept. Suppose that the observed value $y = f(x) + \epsilon$ differs from this function value by an additive noise $\epsilon \sim \mathcal{N}(0, \sigma^2 = 0.25)$ which follows a normal distribution with zero mean, and variance 0.25. If the prior distribution over w is chosen to be a standard normal distribution (zero mean and unit variance), after observing the single datapoint $(x_0, y_0) = (1, 2)$ what is the mean μ_{post} and variance σ^2_{post} of the posterior distribution over w? (up to one decimal place)

$\mu_{post} =$ [ ]

$\sigma^2_{post} =$ [ ]

[short] Bias and Variance

Frage 17 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Which of the following is/are correct?

- [ ] Increasing the capacity of the model decreases the bias
- [ ] Increasing the capacity of the model increases the bias
- [ ] Increasing the capacity of the model increases the variance
- [ ] Increasing the capacity of the model decreases the variance

[short] T/F: AE/GANs

Frage 18 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Select all that apply: Which of the following statements is/are generally true?

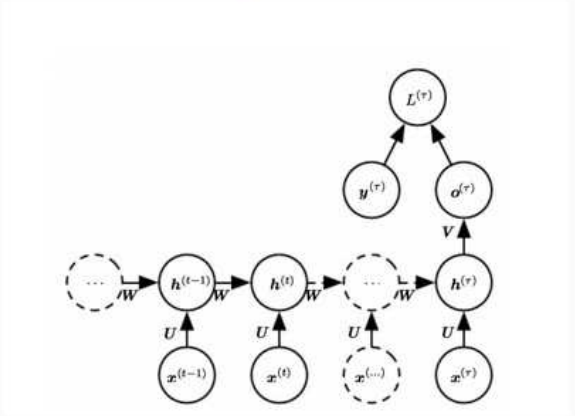
- [ ] The decoder network of a VAE is a generative model
- [ ] The generator network in GANs takes random noise as input
- [ ] GANs are a kind of auto-encoder
- [ ] Overcomplete auto-encoders cannot learn the identity function
- [ ] VAEs add an L1 penalty on the latent code to the loss function

[short] RNN Applications

Frage 19 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

For which of the following tasks is the RNN architecture shown below well-suited?



- [ ] Video Frame Classification
- [ ] Machine Translation
- [ ] Sentiment Analysis
- [ ] Image Captioning

Calculating Self-Attention

Frage 20 von 58 (2 Punkte)
Nicht beantwortet

Aktionen

Consider an attention block using dot-product similarity, having only a single head, and the following weight matrices

```
W^Q = [[0.0], [1.0]]

W^K = [[1.0], [0.0]]

W^V = [[0.5], [0.5]]

W^O = [[0.5, 0.5]]
```

for queries, keys, values, and output respectively.

What would be the output of the block when using it for (unmasked) self-attention on the following input sequence?

```
x = [[1, 1], [1, 1], [0, 0]]
```

Note: No biases are used, the dimensionality for queries/keys/values is 1.

Your answer (rounded to two decimals):

[[,], [,], [,]]

(grading note: 1/3pt per correct entry)

General Architecture of CNNs

Frage 21 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Complete the statements below by selecting the correct options:

Convolutional neural networks typically have a ---- bitte auswählen ---- number of filters close to the input and a ---- bitte auswählen ---- number of filters close to the output (1/3pt).

If you would want to replace a 7x7 convolutional layer, you need ---- bitte auswählen ---- 3x3 convolutional layers with a stride and dilation of 1 stacked on top of each other to have the same receptive field (1/3pt). The stack of 3x3 convolutional layers has ---- bitte auswählen ---- parameters than the 7x7 convolutional layer (1/3pt).

1) larger / smaller 2) larger / smaller 3) 2 / 3 / 4 / 5 / 6 / 7 4) less / more

[short] Memory Complexity Foward Pass

Frage 22 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

The memory complexity of the forward pass of a fully connected neural network with N layers of size K (including the output layer) for a single data point with K features, including the memory needed to store the entire network, is

- ○ $O(NK^2)$
- ○ $O(N)$
- ○ $O(N+K^2)$
- ○ $O(N^2+K)$
- ○ $O(NK)$
- ○ $O(K)$
- ○ $O(N^3)$
- ○ $O(N^2K)$
- ○ $O(\max(N^2,K^2))$
- ○ $O(K^3)$

The Transformer Architecture

Frage 23 von 58 (3 Punkte)
Nicht beantwortet

Aktionen

Consider the original transformer architecture from the "Attention Is All You Need" paper by Vaswani et al (2017).

Furthermore, assume the following architectural parameters:

- **maximum input sequence length: 128**
- **maximum output sequence length: 256**
- **input vocabulary size: 20000**
- **output vocabulary size: 10000**
- **embedding size: 512**
- **number of attention heads: 8**
- **embedding size per head: 64**
- **number of encoder blocks: 6**
- **number of decoder blocks: 6**

Calculate the following quantities (assuming a batch-size of 1):

- The total number of self-attention blocks (0.5pt): [ ] (note: count both masked and unmasked)
- The total number of skip connections (0.5pt): [ ]
- The total number of elements in the output tensor of the last encoder block (0.5pt): [ ]
- The total number of elements in the output tensor of scaled dot product similarity (i.e., the attention weight matrix, for a single head) of an encoder-decoder attention block (0.5pt): [ ]
- The number of output units of the final fully connected layer across which the softmax is calculated (0.5pt): [ ]
- The total number of weights (no biases!) in a single multi-head self-attention module (0.5pt):

- The total number of weights (no biases!) in a single multi-head self-attention module (0.5pt):

[short] Batchnorm: Number of Parameters

Frage 24 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Given input X of dimensions 100 x 12, where 100 is the number of samples and 12 is the dimension of each sample.

How many trainable parameters does a batch normalization layer (in an MLP) have that is trained on this input?

- ○ 1 parameter
- ○ 2 parameters
- ○ 12 parameters
- ○ 24 parameters
- ○ 100 parameters
- ○ 200 parameters
- ○ 1200 parameters
- ○ 2400 parameters

[short] Zero Jacobian

Frage 25 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Select all that apply. The Jacobian is zero in:

- ☐ Local optima
- ☐ Saddle points
- ☐ Points which have all first order derivatives equal to zero
- ☐ Global optima

[short] Advantages of BBO

Frage 26 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Which of the following are advantages of Blackbox optimization?

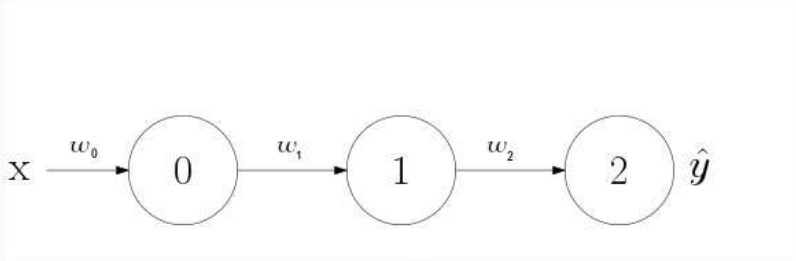
- ☐ Can optimize non-differentiable functions
- ☐ Can be applied to functions not available in closed form
- ☐ Can exploit analytical gradient information
- ☐ Does not suffer from the curse of dimensionality

Backpropagation in MLPs

Frage 27 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Given the neural network described below, calculate the derivative of the loss w.r.t. w_0.



Consider the following initial values for your calculations:

- **x = 20**
- **y = −3**
- **$w_0 = w_1 = -0.5$**
- **$w_2 = 0.5$.**

The nonlinearities for the first two units are PReLU with the same parameter a = 0.1. We do not use a nonlinearity for the last unit: $g_2(z_2) = z_2 = \hat{y}$.

There are no biases and we use L1-loss.

The forward pass results are given:

- **Preactivation $z_0 = -10$**
- **Activation $g_0 = -1$**
- **Preactivation $z_1 = 0.5$**
- **Activation $g_1 = 0.5$**
- **Preactivation $z_2 = 0.25$**
- **Activation $g_2 = 0.25$**
- **The output $\hat{y} = g_2 = 0.25$**
- **The loss $L = |\hat{y} - y| = 3.25$**

[short] Measures against underfitting

Frage 28 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Which of the following measures can we take against underfitting?

- [ ] Increase representational capacity
- [ ] Tune the optimizer's hyperparameters
- [ ] Increase data augmentation
- [ ] Increase dropout rate

[short] Bayesian Terminology

Frage 29 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Given weights w, a design matrix X and labels y and the bayes formula as given below, name each term (order matters for the multiplication in the numerator!)

$$p(\boldsymbol{w}|\boldsymbol{y}, \boldsymbol{X}) = \frac{p(\boldsymbol{y}|\boldsymbol{X}, \boldsymbol{w})p(\boldsymbol{w})}{p(\boldsymbol{y}|\boldsymbol{X})}$$

---- bitte auswählen ---- = (---- bitte auswählen ---- * ---- bitte auswählen ----)/
---- bitte auswählen ----

1) posterior / prior / marginal likelihood / likelihood
2) posterior / prior / marginal likelihood / likelihood
3) posterior / prior / marginal likelihood / likelihood
4) posterior / prior / marginal likelihood / likelihood

ELBO

Frage 30 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

$$KL(q_\phi(\boldsymbol{\theta})||p(\boldsymbol{\theta}|\mathcal{D})) + \underbrace{\mathbb{E}_{q_\phi(\boldsymbol{\theta})}\left[\log p(\mathcal{D}|\boldsymbol{\theta})\right] - KL(q_\phi(\boldsymbol{\theta})||p(\boldsymbol{\theta}))}_{\text{evidence lower bound (ELBO)}} = \log p(\mathcal{D})$$

Briefly answer the following questions about the evidence lower bound (ELBO):

1. **Which term do we actually want to optimize when maximizing the ELBO?**
2. **Do we want to optimize or maximize this term?**
3. **Why do we not optimize this term directly?**
4. **Where is the 'evidence' in this formula?**
5. **Why is ELBO a lower bound on the evidence?**

B I

P

[short] Parameter Sharing

Frage 31 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Select all that apply: Which of the following statements is/are generally true about parameter sharing?

- [ ] Parameter sharing reduces the memory required to store a model
- [ ] Parameter sharing forces weights to zero
- [ ] RNNs apply parameter sharing across time
- [ ] Parameter sharing lowers the representational capacity of a model

[short] History of Deep Learning

Frage 32 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Order the following milestones of deep learning chronologically from early (top) to late (bottom).

Development of Autoencoders (Baldi and Hornik)

Introduction of backpropagation algorithm (Rumelhart et al.)

Greedy layer-wise pretraining (Hinton et al.)

First version of stochastic gradient descent (Widrow and Hoff)

- [ ] Die vorgegebene Anordnung ist korrekt.

[short] LSTM-GRU

Frage 33 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Which of the following statements is/are true?

- [ ] The number of trainable parameters in a sequence model with LSTM units grows linearly with the sequence length.
- [ ] GRUs use a single gating unit to control the forgetting factor and therefore cannot capture long term dependencies.
- [ ] LSTMs have more trainable parameters than GRUs with the same input and hidden size.
- [ ] In a long sequence model with LSTM cells, if gradient clipping is not used, the values of the hidden state entries may grow unboundedly during training

[short] Decoupled Weight Decay

Frage 34 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Construct a true statement about decoupled weight decay:

Using ---- bitte auswählen ---- with decoupled weight decay is equivalent to using ---- bitte auswählen ---- with ---- bitte auswählen ---- regularization (for some λ and a potentially different learning rate).

1) Adam / SGD 2) SGD / Adam 3) L1 / L2

Debug Training Pipeline

Frage 35 von 58 (2 Punkte)
Nicht beantwortet

Aktionen

The training pipeline given below contains two errors. These errors are present inside the todo-blocks.

Describe what needs to be changed for the pipeline to be functional.

Note: Make sure to be specific. Ideally, provide the corrected code.

```
import torch
import torch.nn as nn
import torch.optim as optim

trainloader = torch.utils.data.DataLoader(...)
num_epochs = 10
net = nn.Sequential(...)

### START TODO ###
criterion = nn.CrossEntropyLoss()
optimizer = optim.Adam(lr=0.001)

for epoch in range(num_epochs):
    for inputs, labels in trainloader:
        outputs = net(inputs)
        loss = criterion(outputs, labels)
        loss.backward()
        optimizer.step()
### END TODO ###

torch.save(net.state_dict(), 'model_weights.pth')
```

[short] Dimensionality of the Gradient

Frage 36 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

The dimensionality of the gradient of the loss w.r.t. the incoming weights of all neurons in a layer of size k with n incoming connections per neuron is:

- ○ A vector of dimension k x 1
- ○ A matrix of dimension n x k
- ○ A scalar
- ○ A vector of dimension n x 1

[short] Concepts of Backpropagation

Frage 37 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Which of the following statements is/are correct?

- ☐ In MLPs, backpropagation has the same asymptotic (Big O) time complexity as inference.
- ☐ We cannot use torch.max/torch.min operator in a neural network, if we want to perform backpropagation.
- ☐ Because the Mean Squared Error loss is convex, theoretically, the neural network trained with MSE loss will converge to the global optimum.
- ☐ In order to perform backpropagation, we need to store the output value of each hidden layer.

Data Augmentation: Reflection

Frage 38 von 58 (2 Punkte)
Nicht beantwortet

Aktionen

When should you use reflection (a.k.a. mirroring or flipping) in image processing applications?

1. **Give an example of an application where reflection is a reasonable augmentation [1pt].**
2. **Give an example of an application where reflection is a particularly bad one and briefly explain (in 1-2 sentences) why it should not be used [1pt].**

P

[short] Receptive Field

Frage 39 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Select all that apply: The Local Receptive Field of a neuron:

- [ ] Decreases with depth of the convolutional neural network
- [ ] Depends on the sizes of the kernels of the convolutions
- [ ] Increases with depth of the convolutional neural network

Configuration Space

Frage 40 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

How many different configurations does the following configuration space contain?

Hyperparameter	Choices	Default	Conditions
optimizer	["SGD", "Adam"]	"SGD"	None
sgd_momentum	[0.2, 0.5, 0.9]	0.9	Active only if optimizer="SGD"
adam_beta1	[0.8, 0.9, 0.99]	0.9	Active only if optimizer="Adam"
adam_beta2	[0.9, 0.99, 0.999]	0.999	Active only if optimizer="Adam"
weight_decay	[0.0, 0.1, 0.2, 0.3]	0.0	None
lr	[0.01, 0.001, 0.0001]	0.0001	Active only if weight_decay>0.0

[short] Training of GANs

Frage 41 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

When training GANs...

- ○ You first train the generator, then the discriminator, done!
- ○ You train discriminator and generator alternatingly
- ○ You only train the generator
- ○ You first train the discriminator, then the generator, done!
- ○ You only train the discriminator

[short] Definition of an Activation Function

Frage 42 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

How is the Logistic hyperbolic tangent function defined? Select all correct answer(s)

- ☐ $f(x) = (e^{x} - e^{-x}) / e^{-x}$
- ☐ $f(x) = (e^{x} - e^{-x}) / (e^{x} + e^{-x})$
- ☐ $f(x) = (e^{x} - e^{-x}) / e^{x}$
- ☐ $f(x) = (e^{-x} - e^{x}) / (e^{x} + e^{-x})$

Backward Pass CNN

Frage 43 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Perform a single forward pass and backward pass of this 1D cross-correlation with

- **Stride 2, no padding**
- **Learning rate, α = 0.01**
- **MSE Loss**
- **SGD optimizer**

on the following input (a single 1D image):

$$x = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

use the following weights and bias for the 1D filter,

$$w = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$b = 1$$

and as ground truth (i.e. the target output of the cross-correlation),

$$y = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

What is the updated weight w_1 at the end of the first backward pass?

[ ]

[short] Kinds of Auto-Encoders

Frage 44 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Match the different kinds of auto-encoders to the description of their core characteristic.

(Hint: To answer, drag the auto-encoder name to their matching characteristic)

(grading note: 0.2pt per correct match)

Penalizes the absolute value of the latent code to learn representations with low effective dimensionality	Denoising Auto-Encoder
Penalizes the absolute value of the derivatives of hidden layer activations w.r.t. the input to learn representations that are robust to small changes in the input	Bottleneck Auto-Encoder
The encoder outputs the posterior distribution over the latent variables	Sparse Auto-Encoder
Corrupts the input and learns to reconstruct the original input	Contractive Auto-Encoder
Learns latent codes that are shorter than the input by limiting the number of units in the hidden layer	Variational Auto-Encoder

Adaptive Gradient Method

Frage 45 von 58 (2 Punkte)
Nicht beantwortet

Aktionen

Answer the following questions:

1. **Name three adaptive gradient methods you have learned about in this course [1pt]**
2. **Briefly explain what the main problem is these optimizers try to address [1pt]**

p

[short] Basics of Deep Learning

Frage 46 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Mark the true statements:

- [ ] Deep Learning can handle cases where input and output modality differ (e.g. images as input and text as output).
- [ ] In contrast to other machine learning models, deep learning models are more interpretable.
- [ ] As long as a neural network is deep enough, it is guaranteed to fit to any continuous function, if sigmoid activations are used and it is trained for enough time.
- [ ] Deep Learning generally needs less preprocessing than other Machine Learning models
- [ ] Neural Networks are so powerful because they are structured and learn like the brain.
- [ ] Cross-validation is not needed for neural networks because they do not overfit.

[short] Segmentation Types

Frage 47 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Match the description of different types of segmentation to their respective term.
(to answer: drag the term to the associated description)

Not every pixel is labeled; Pixels of the same class can have different labels.	Instance Segmentation
Every pixel is labeled; The labels represent a class membership.	Panoptic Segmentation
Every pixel is labeled; Pixels of the same class can have different labels.	Semantic Segmentation

Parameter Count CNN

Frage 48 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Order the following layers according to their number of parameters (Higher parameter count at top). If there is a tie, break the tie by ordering the layer higher that does more multiplications given an input tensor of (Batch_Size, 3, 64, 64).

- nn.Conv2d(in_channels=3, out_channels=5, kernel_size=3, stride=1, padding="same", bias=False)
- nn.Conv2d(in_channels=3, out_channels=3, kernel_size=3, stride=2, padding=1, bias=False)
- nn.Conv2d(in_channels=3, out_channels=3, kernel_size=5, stride=1, padding=2, bias=False)
- nn.Conv2d(in_channels=3, out_channels=5, kernel_size=3, stride=1, padding="valid", bias=False)
- nn.Conv2d(in_channels=3, out_channels=3, kernel_size=3, stride=1, padding=1, bias=False)

☐ Die vorgegebene Anordnung ist korrekt.

Number of Parameters in a MLP

Frage 49 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

How many trainable parameters does the neural network defined below have?

```
net = nn.Sequential(
    nn.Linear(10, 20),
    nn.Sigmoid(),
    nn.Linear(20, 8),
    nn.Sigmoid()
)
```

Note: nn is the PyTorch torch.nn module

Hint: Do not forget the biases!

LSTM Module

Frage 50 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Consider the following LSTM module:

```
lstm_net = nn.LSTM(input_size=256, hidden_size=512, bias=False)
```

Calculate the following quantities:

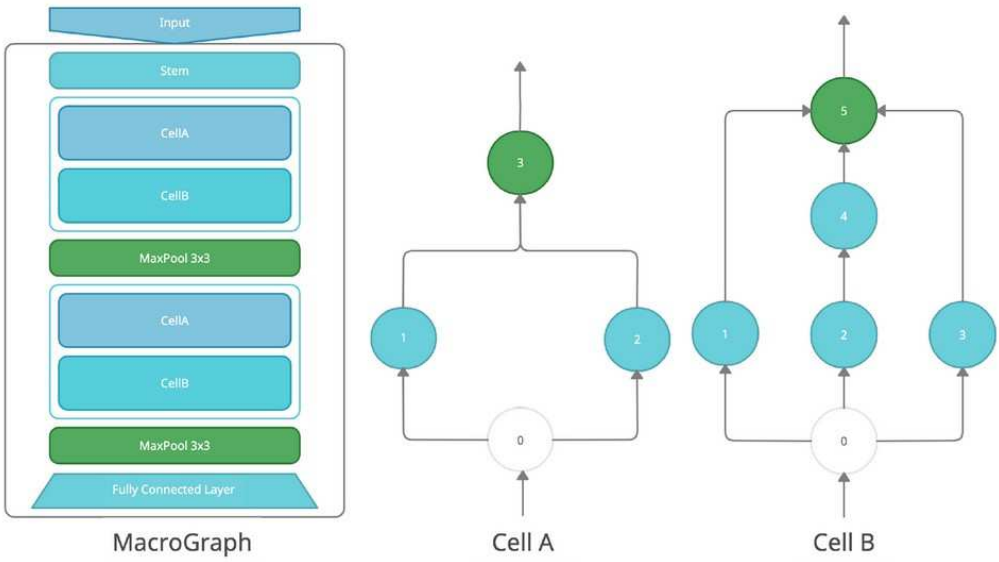
- The dimensionality (size) of the cell state (0.5pt): _____
- The total number of parameters of the LSTM module (0.5pt): _____ (hint: No biases!)

Size of Cell Search Space

Frage 51 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

You are given the following Cell Search space and Macro architecture.



The larger Macro architecture uses CellA and CellB as part of it.

The blue nodes in Cell A and Cell B can contain one of these operations:

- **1x1 convolution**
- **3x3 convolution**
- **5x5 convolution**

followed by one of these activations:

- **ReLU**
- **GELU**

The green and white nodes in Cell A and B are merge layers and identity operators respectively. They do not change.

How many architectures does this search space span?

Simply count the number of possible assignments of operations to nodes here and ignore the fact that e.g. cell symmetry makes some of these architectures equivalent.

_____

MLE, Entropy, Cross-Entropy, and KL divergence

Frage 52 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Let
< br />- d(x) be the likelihood of the data under the true, unknown data distribution.
< br />- p(x) the likelihood of the data under the empirical data distribution
< br />- q(x;w) be the likelihood of the data under the model with parameters w.
< br />
< br />Complete the following sentences to form correct statements (grading: 0.25pt per fully correct statement).
< br />(note: multiple options may be correct)

The maximum likelihood estimator [---- bitte auswählen ----] the cross-entropy between [---- bitte auswählen ----] and [---- bitte auswählen ----].

The maximum likelihood estimator [---- bitte auswählen ----] the Kullback-Leibler divergence D_KL([---- bitte auswählen ----] || [---- bitte auswählen ----]).

The cross-entropy between p and d is [---- bitte auswählen ----] bound on D_KL([---- bitte auswählen ----] || [---- bitte auswählen ----]).

When D_KL(d(x) || q(x;w)) is zero, the cross-entropy between d(x) and q(x;w) is [---- bitte auswählen ----]

1) maximizes / minimizes 2) p(x) / d(x) / q(x,w) 3) p(x) / d(x) / q(x,w)

4) maximizes / minimizes 5) p(x) / d(x) / q(x,w) 6) p(x) / d(x) / q(x,w)

7) an upper / a lower 8) p(x) / d(x) / q(x,w) 9) p(x) / d(x) / q(x,w)

10) equal to the entropy of q(x,w) / equal to the entropy of d(x) / equal to the entropy of p(x) / smaller than zero / zero / smaller or equal to 0

[short] First NAS Approach

Frage 53 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Which of the following is historically the first NAS approach?

- ○ Differentiable Architecture Search
- ○ Reinforcement Learning
- ○ Neuroevolution
- ○ Bayesian Optimization

MCMC sampling

Frage 54 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Consider a state space consisting of all integers $\mathbb{Z} = \{\cdots, -2, -1, 0, 1, 2, \cdots\}$, and suppose the target distribution and the proposal distribution over this space are given respectively by $P_{target}(i) \propto e^{-|i|}$ and $g(i|j) \propto e^{-2(i-j)^2}$ up to a normalization constant, where $i, j \in \mathbb{Z}$. Starting from the state $i = 1$, and after one step of Metropolis-Hastings MCMC sampling, let $P(2|1)$ and $P(0|1)$ denote the probability of ending up in state 2 and 0 respectively. What is the ratio $\frac{P(2|1)}{P(0|1)}$?

- ○ e^3
- ○ e^2
- ○ e
- ○ 1
- ○ 1/e
- ○ 1/e^2
- ○ 1/e^3
- ○ 0

3D Convolutions

Frage 55 von 58 (2 Punkte)
Nicht beantwortet

Aktionen

Consider applying the following 3D convolution on an input volume 8 x 3 x 100 x 32 x 32 (B x C x D x H x W)

```
conv = nn.Conv3d(3, 10, kernel_size=(5, 5, 5), bias=False)
```

Note:

- **nn here is the torch.nn module of PyTorch.**
- **Don't forget that bias=False in the given code!**

Calculate the following quantities:

- the number of parameters of the 3D convolution: [ ] (0.5pt)
- the number of multiplications performed: [ ] (1pt)
- the output volume (B x C x D x H x W): [ ] x [ ] x [ ] x [ ] x [ ] (5 x 0.1pt)

[short] Goal Regularization

Frage 56 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

The ultimate goal of regularization is ...

- ○ increasing the sparsity of neural networks
- ○ reducing the generalization error
- ○ reducing the size of neural networks
- ○ reducing memory costs
- ○ reducing bias
- ○ reducing the training loss
- ○ counter-acting underfitting
- ○ reducing the validation loss
- ○ speeding up model training

[short] Preconditioning Optimizers

Frage 57 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

Which of these optimizers use preconditioning?

- ☐ SGD
- ☐ Adagrad
- ☐ Adam
- ☐ RMSProp

Multiplications in FC Layers

Frage 58 von 58 (1 Punkt)
Nicht beantwortet

Aktionen

How many multiplications are performed in the forward pass of a batch of 16 data points, each having 784 features, through a fully connected layer with 120 units (without biases)?